heap’s leaf and push it down, because we need to keep into account the BST’s constraints as well.

Likewise, when we insert (or delete) a node, we can’t just use the simple BST algorithm: if we just search for the position that the new key would hold in the tree and it as a leaf, as shown in figure 3.3, the BST constraint will still be abided by, but the new node’s priority might violate the heap’s invariants.

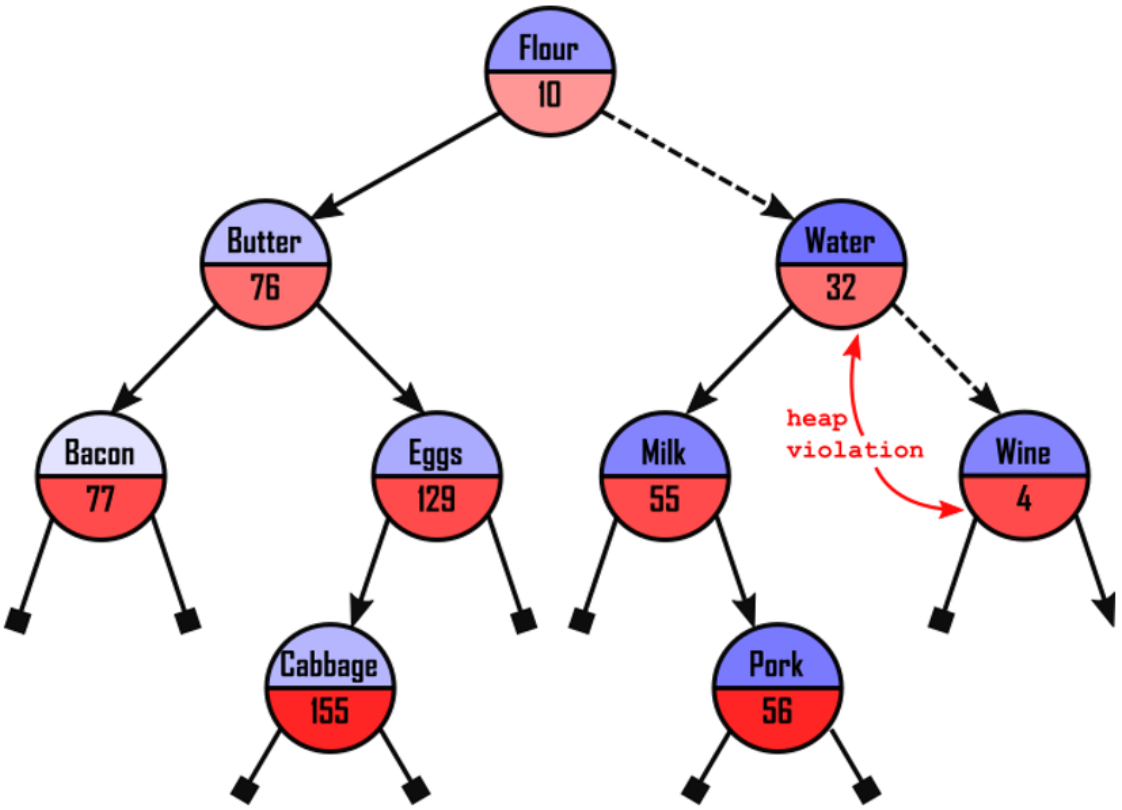


Figure 3.3 An example of the insertion of a node in a treap, based on the key’s ordering only: the priority of the new node, however, violates the heap constraints.

Listing 3.1 introduces a possible implementation for the treap’s main structure; we will use an auxiliary class to model tree’s nodes, and this will be instrumental in our implementation. You might have noticed we are using explicit links to node’s children,